Identifying Plant Pathogens with a CNN-Based Mode

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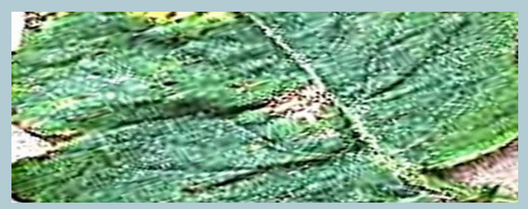
Introduction

Plant pathogens pose a threat to global food security and the health of ecosystems. This investigation developed a Convolutional Neural Network (CNN) model to identify common plant pathogens based on images of leaves.

CNNs were used in this study to identify plant diseases using a publicly available dataset containing images of healthy and diseased plant leaves across multiple disease categories. It was based on 5 classes: Healthy, \, Fungi, Pests, and Virus. The model was trained by testing its recall, precision, accuracy, and a confusion matrix. On the first trial, the model was able to achieve a precision of 95% and an accuracy of 94%. After running the model through a second trial, the results surged. The model reached an accuracy of 97% and a precision of 98%. The confusion matrix revealed that the model excelled at identifying powdery mildew (98% precision), while it often confused early blight with leaf spot diseases.

Like in previous research, this model was able to achieve high values of performance that can be a powerful tool for plant disease detection. However, this model is different from others since it does not rely on handcrafted features; the CNN model can learn directly from the raw images. This makes the model easier to apply to different data sets. Based on its performance, this model shows enough evidence to prove that it can be an efficient solution for farmers and other individuals who work in the environmental sector.



-Ex. Image of Bacteria used for testing

Purpose

Plant diseases are a threat to food security, affecting agriculture and causing significant economic losses. Finding a time efficient solution is needed to provide effective crop management. Traditional strategies- such as running visual tests by experts- can be time consuming, expensive, and can be impracticable for farmers.

Artificial intelligence and programming languages have made the process of plant disease detection easier to manage. In this project, we use Python to create a machine learning model, that can identify plant pathogens from leaf images.

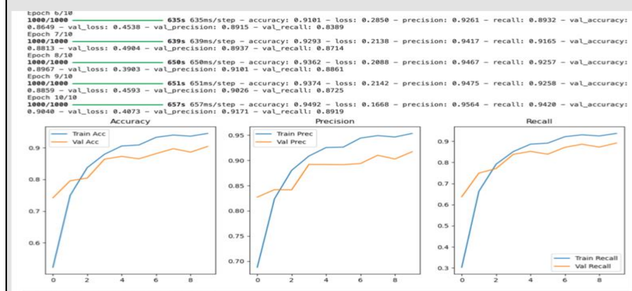
This method provides a time efficient and accessible solution for diagnosing plant disease. Our goal is to explore the potential that artificial intelligence has in agriculture and show how these tools can be useful to improve food systems.

Methods

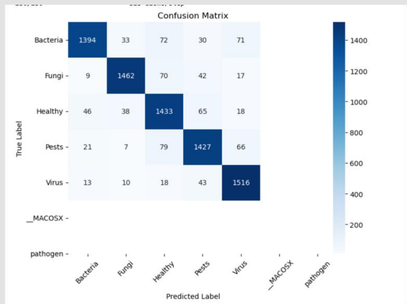
The way a convolutional network works is that it undergoes a series of process called pooling where they extract information from the matrix created. These are the different processes that the convolutional network model went through to identify the plant pathogens:

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 222, 222, 32)	896
max_pooling2d_3 (MaxPooling2D)	(None, 111, 111, 32)	0
conv2d_4 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_4 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_5 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_5 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten_1 (Flatten)	(None, 86528)	0
dense_2 (Dense)	(None, 512)	44,302,848
dense_3 (Dense)	(None, 7)	3,591

After going through these processes, the model was ready to start training.
It was tested twenty times to it get used to the matrix. Following the training, there was graphs created to test how accurate these model was. These was the information gained from the first training of the model:



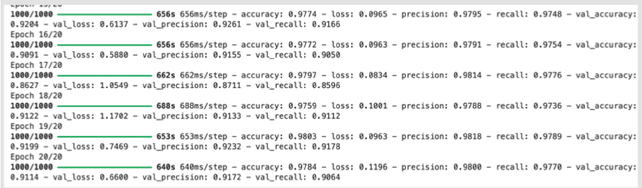
prominent. The graphs show that the training models surpassed the actual value of the model. These indicates that the model is not as accurate as it is shown. Therefore, after running these processes, we ran the model through a confusion matrix, where we were able to tests its performance accurately.



The confusion matrix shows that the model is accurate. The dark blue boxes show the correct predictions which constitutes a major part of the tested variables. After analyzing the confusion matrix, it reveals that the model is ready to be tested again to improve its performance.

Results

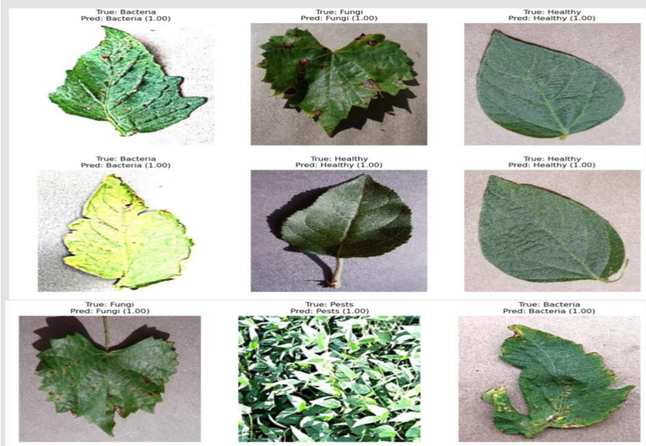
After the second trial of training, the result improved . The model reached its highest state, it was getting used to the matrix. These are the new results:



It was able to obtain an accuracy of 97%, precision of 98%, and a recall of 97%. Because of these results, it was clear that the model was ready to be tested. At the start, I only trained it based on one class, which was bacteria. These was the first set of training:

The model was able to predict the plant pathogen from each image. However, as the model was exposed to the images, it did get some of them wrong. For example, in the bacteria class, it was confused with pests for one of the images:

Following this trial of testing, the other classes were included. This was intended to see the accuracy of the model when identifying multiple plant pathogens. When testing multiple pathogens, the model predicted more accurately than the trial where only one pathogen was tested. These were the of the trial including all pathogens:



Discussion

Compared to previous plant disease detection projects, this one differs due to its time efficiency and accessibility. It is especially useful for farmers that are in settings that do not have such technological artifacts. Even though initiatives rely on equipment, this project uses machine learning to create a model that can be run by any farmer. Visual inspections, which can be costly, are not needed for this process. It diagnoses images through a pooling process by detecting information of the plant diseases from its matrix. Through its focus on improving food systems, AI can offer efforts to global efforts to find farmer –friendly solution for these problems.

Conclusion

This project demonstrates the potential of convolutional neural networks in the environmental field, specifically identifying plant pathogens through analysis by images. By using deep learning networks, we were able to develop a model that was capable of classifying plant diseases from leaf images with high efficiency. This method works around the limitations of traditional methods used to diagnose , that rely on experts to visualize the plants. The traditional methods can be time consuming, and costly.

The CNN model can enable the use of tools throughout the environmental sector. Throughout the project, it is show how machine learning can be trained based on image data sets. These data sets can adapt to the model and provide effective solutions. As it is shown by the performances, this model can be relied on as a future tool in the environmental field.

References

1. deeplizard. "Convolutional Neural Networks (CNNs) Explained." YouTube, 9 Dec. 2017, www.youtube.com/watch?v=YRhxdVk_sIs
2. Optimistic, An. "Convolutional Neural Networks Explained (CNN Visualized)." YouTube, www.youtube.com/watch?v=pj9-rr1wDh19 Dec. 2020, M.
3. IBM. "Convolutional Neural Networks." Ibm.com, 6 Oct. 2021, www.ibm.com/think/topics/convolutional-neural-networks.

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